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Your Roll No.....

Sr. No. of Question Paper : 1566

F-8

Unique Paper Code : 2351402

Name of the Paper : Algebra – III (Ring Theory and Linear Algebra – I)

Name of the Course : B.Sc. (Hons.) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any two parts from each question.

1. (a) Let $M_2(\mathbb{Z})$ be the ring of all 2×2 matrices over $\mathbb{Z}$, the set of all integers and

$$R = \left\{ \begin{bmatrix} a & a+b \\ a+b & b \end{bmatrix} : a, b \in \mathbb{Z} \right\}.$$

Prove or disprove that R is a subring of $M_2(\mathbb{Z})$.

- (b) Define an idempotent element of a ring. Show that in a commutative ring of characteristic 2, the idempotents form a subring.

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- (c) Let R be a ring with unity 1 . If 1 has order n under addition, then prove that the characteristic of R is n .
2. (a) Show that $\mathbb{R}[x]/\langle x^2 + 1 \rangle$ is a field.
- (b) Define Prime ideal and Maximal ideal of a ring with unity. Prove that every maximal ideal is prime in a commutative ring with unity. Is the converse true? Justify.
- (c) Determine all the distinct elements of the quotient ring R/I , where $R = M_2(\mathbb{Z})$, the ring of all 2×2 matrices over $\mathbb{Z}$ and I is the ideal of R consisting of matrices with even entries.
3. (a) Determine all ring homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}$.
- (b) State and prove the First Isomorphism Theorem for rings.
- (c) Show that a homomorphism from a field to a ring with more than one element must be an isomorphism onto its image.
4. (a) Prove that the Span of any subset S of a vector space V is a subspace of V . Moreover, any subspace that contains S must also contain the Span of S .

- (b) Let V be a vector space over a field of characteristic not equal to 2. Let u, v and w be distinct vectors in V . Prove that $\{u, v, w\}$ is linearly independent if and only $\{u + v, u + w, v + w\}$ is linearly independent.
- (c) The vectors $\{(2, -3, 1), (1, 4, -2), (-8, 12, -4), (1, 37, -17), (-3, -5, 8)\}$ generate $\mathbb{R}^3$. Find a subset of this set that is a basis for $\mathbb{R}^3$. . . (6.5,6.5,6.5)
5. (a) Let V and W be finite dimensional vector spaces over a field F and $T: V \rightarrow W$ be a linear transformation. Prove that $\text{nullity}(T) + \text{rank}(T) = \dim V$.
- (b) Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $T(a_1, a_2) = (a_1 - a_2, a_1, 2a_1 + a_2)$. Let β be the standard ordered basis for $\mathbb{R}^2$ and $\gamma = \{(1, 1, 0), (0, 1, 1), (2, 2, 3)\}$. Compute $[T]_{\beta}^{\gamma}$.
- (c) Let V be a vector space and let $T: V \rightarrow V$ be linear. Prove that $T^2 = T_0$ if and only if $R(T) \subseteq N(T)$. (6.5,6.5,6.5)
6. (a) Let V and W be finite dimensional vector spaces over the same field F . Prove that V is isomorphic to W if and only if $\dim V = \dim W$.
- (b) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x_1, x_2, x_3) = (3x_1 - 2x_3, x_2, 3x_1 + 4x_2)$. Prove that T is invertible and find $T^{-1}(x_1, x_2, x_3)$.

(c) Let T be the linear operator

$$T \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2a + b \\ a - 3b \end{bmatrix}. \text{ Let } \beta \text{ be the sta}$$

for $\mathbb{R}^2$ and let $\beta' = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$. Find

Q such that $[T]_{\beta'} = Q^{-1}[T]_{\beta}Q$.